

Goal-Instrumental Constraint Neglect in LLM Assistants

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Abstract

Instruction-following evaluations usually treat user constraints as requirements to satisfy or as directly incompatible instructions to resolve, but a constraint can also defeat the user’s own stated goal. We study this goal-instrumental constraint neglect setting with GICN, a relation-label diagnostic that asks whether a stated constraint supports, blocks, or is orthogonal to the user’s goal. Using Qwen2.5-7B-Instruct served with vLLM under deterministic decoding, we compare direct answering, a generic conflict-check prompt, a structured audit wrapper, and wrapper ablations on 40 paired GICN rows plus selected public-control slices. Direct answering reaches 0.700 relation-label accuracy, and the generic conflict-check prompt reaches 0.800. The audit wrapper reaches 0.975, and the no-feasible-alternative ablation reaches 0.925. The result supports a bounded diagnostic claim: the repaired wrapper improves the GICN relation decision by reducing compatible-control false alarms while preserving trap detection, but the five-row control slices and single evaluated configuration do not support broad no-harm or model-family generality claims.

Introduction

Instruction-following benchmarks have made literal constraint adherence measurable; IFEval introduced verifiable instruction-following prompts (Zhou et al. 2023). Follow-Bench, InFoBench, and ComplexBench push that idea toward fine-grained, decomposed, and composed constraints (Jiang et al. 2024; Qin et al. 2024; Wen et al. 2024). Conflict-oriented benchmarks move closer to the setting studied here by asking models to detect or resolve incompatible instructions (He et al. 2025; Javed et al. 2026). These lines of work leave a narrower failure mode: the user may specify a goal and a constraint, and the constraint may be instrumentally related to the goal in a way that makes literal obedience counterproductive. A model that preserves the words of the constraint can still fail the user’s end.

We call this failure mode goal-instrumental constraint neglect. The central question is not only whether a model can follow a constraint, nor only whether two instructions are syntactically incompatible. The question is whether a model can identify a means-end relation: does the stated constraint support the goal, block the goal, or remain orthogonal to the goal? A toy example is a request to wash a car while also requiring that the car not get wet. The important behavior

is recognizing that the constraint blocks the goal and asking for clarification or a feasible relaxation, while avoiding false alarms on compatible constraints.

This paper reports a bounded diagnostic study rather than a broad mitigation-success claim. We build a repaired GICN diagnostic with trap and compatible rows, evaluate train-free condition families, and include selected public instruction-following control slices. The evaluated condition families are direct answering, a generic conflict-check prompt, a full audit wrapper, and ablations of that wrapper. Figure 1 summarizes the diagnostic flow: a user goal and stated constraint are mapped to a relation label, each prompt condition is evaluated with the same instruction-tuned model/backend, and a verifier compares predicted labels with gold labels.

The contribution is a scoped diagnostic argument. GICN makes goal-instrumental relations explicit and distinguishes blocking constraints from compatible controls. The 40-row diagnostic and selected controls evaluate one instruction-tuned model/backend under train-free prompt conditions. The central empirical result is a bounded positive diagnostic result: direct answering reaches 0.700 and generic conflict check reaches 0.800. The full audit wrapper reaches 0.975, and the compatible-examples ablation also reaches 0.975. The audit wrapper’s gain over generic conflict check is driven by compatible-control rows while both conditions preserve perfect trap detection, so the supported story is false-alarm reduction on this diagnostic, not broad deployment-ready mitigation.

The distinction is practical for evaluation design. A benchmark that asks only whether the assistant followed explicit constraints can mark literal obedience as success even when the constraint defeats the user’s goal. A benchmark that asks only whether two instructions conflict can miss the user-goal relation when the conflict is not a direct pairwise contradiction. GICN therefore treats the goal as the anchor and asks whether the stated constraint is instrumentally helpful, harmful, or irrelevant. This is a small target, but it is a target that existing instruction-following and conflict benchmarks do not isolate cleanly.

This paper also treats evidence gaps as part of the result. The current run supports the wrapper on the repaired GICN diagnostic, but it does not support broad no-harm behavior because each public-control family contributes only five selected rows. The paper should therefore be read as a diag-

Goal-Instrumental Constraint Diagnostic Overview

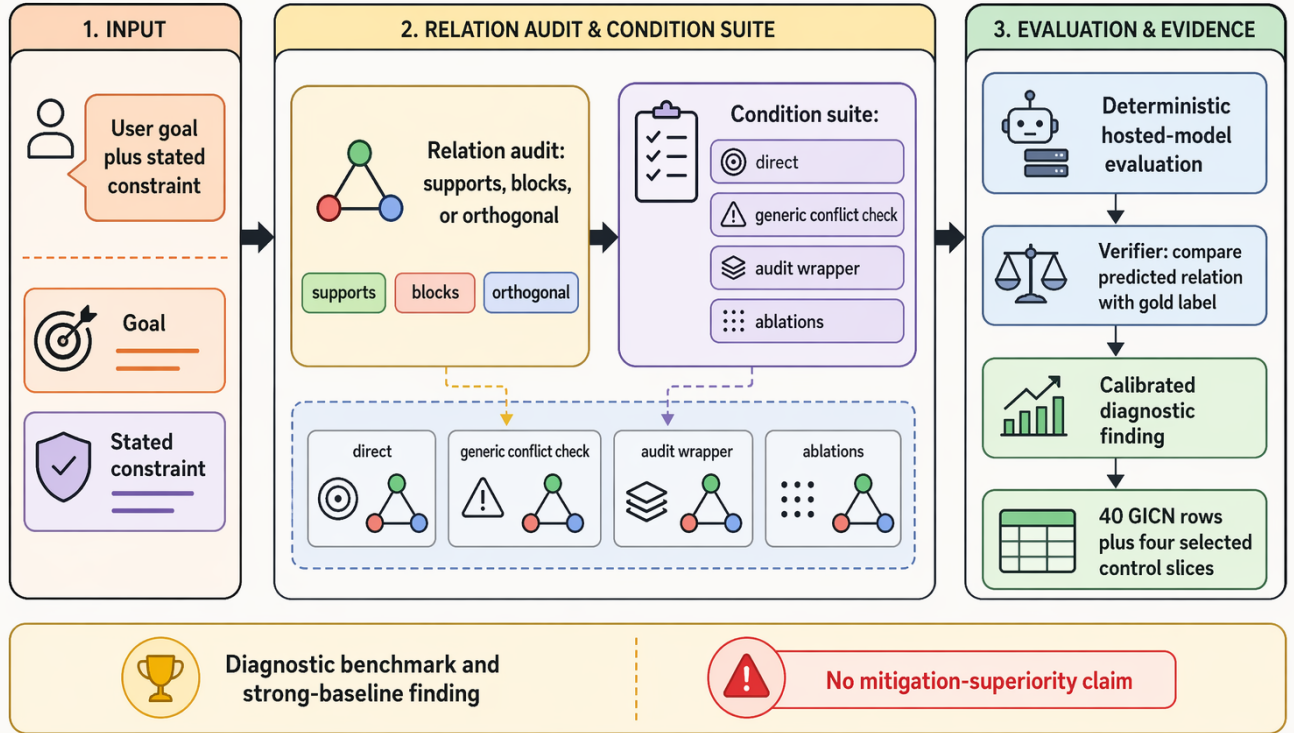


Figure 1: Figure 1 gives the paper-facing GICN protocol. A user goal and stated constraint define a relation-label diagnostic; the condition suite evaluates direct answering, a generic conflict check, the audit wrapper, and ablations on the same rows. The inference backend produces predictions, and the verifier compares each predicted relation against the gold label. The figure records the bounded-scope caveat because the current evidence supports GICN diagnostic improvement, not broad mitigation or no-harm generality.

nostic benchmark report with a calibrated positive result: structured auditing reduces GICN compatible-control false alarms relative to generic conflict checking while preserving trap detection. The rest of the paper is organized around that bounded interpretation.

Related Work

Instruction following and constraint benchmarks. Literal instruction-following evaluation has advanced through increasingly fine-grained constraint measurements. IFEval introduced verifiable prompts for instruction following (Zhou et al. 2023); FollowBench scores fine-grained instruction levels (Jiang et al. 2024); InFoBench decomposes instructions into checkable requirements (Qin et al. 2024); and ComplexBench stresses multiple-constraint composition (Wen et al. 2024). Work on complex-to-simple prompting and constraint back-translation further improves or probes multi-constraint instruction following (He et al. 2024; Qi et al. 2025). These benchmarks motivate the controls used here, but they mainly treat constraints as obliga-

tions. GICN instead asks when a user-stated constraint itself blocks the user’s goal.

Conflict detection and incompatible instructions. ConInstruct and PRIME directly study conflict detection, resolution, or incompatible prompt behavior (He et al. 2025; Javed et al. 2026). MulDimIF adds a multi-dimensional constraint taxonomy and filters compatible constraint-rich instructions for evaluation and training (Ye et al. 2026). These works are closest to the present setting and define the main novelty risk. GICN is narrower: it does not ask whether two instructions conflict in the abstract, but whether a constraint has a supporting, blocking, or orthogonal relation to the goal that the same user stated.

Train-free prompting, self-feedback, and agent loops. Prompting and agent work shows that structured reasoning, self-feedback, and tool-augmented loops can improve task behavior without changing model weights (Yao et al. 2023b; Shinn et al. 2023; Madaan et al. 2023; Schick et al. 2023). Tool-oriented systems and benchmarks such as ToolLLM, API-Bank, Gorilla, HuggingGPT, and MRKL systems frame

Table 1: GICN relation labels separate helpful constraints from blocking constraints; the 40-row diagnostic contains 20 compatible controls and 20 blocking traps.

Label	Diagnostic meaning	Scoring risk
Supports	constraint helps the goal	false block alarm
Blocks	constraint prevents the goal	missed conflict
Orthogonal	constraint is irrelevant	spurious warning

external action and modular reasoning as first-class components (Qin et al. 2023; Li et al. 2023; Patil et al. 2023; Shen et al. 2023; Karpas et al. 2022). Our audit wrapper is in this train-free family, but the current evidence does not justify presenting it as the winning method.

Evaluation, judges, and agent benchmarks. LLM-as-judge and broad evaluation work provides context for model-based assessment and benchmark design (Zheng et al. 2023; Liu et al. 2023; Liang et al. 2023). Truthfulness and hallucination benchmarks show the value of targeted behavioral diagnostics (Lin, Hilton, and Evans 2022; Manakul, Liusie, and Gales 2023). AgentBench, WebArena, GAIA, SWE-bench, ALFWorld, WebShop, Voyager, MemGPT, and Generative Agents illustrate the broader movement toward evaluating assistants in interactive or long-horizon settings (Liu et al. 2025; Zhou et al. 2024; Mialon et al. 2023; Jimenez et al. 2024; Shridhar et al. 2021; Yao et al. 2023a; Wang et al. 2023; Packer et al. 2024; Park et al. 2023). Process-supervision and bootstrapped reasoning work similarly motivates separating what the model says from what the evaluator can verify (Zelikman et al. 2022; Lightman et al. 2023). GICN contributes a compact relation-label diagnostic inside this larger evaluation landscape.

The literature also motivates the conservative scope of the present evidence. Public instruction-following benchmarks are valuable controls because they expose whether an audit prompt breaks ordinary constraint satisfaction, but they do not by themselves validate goal-instrumental reasoning. Conversely, conflict-detection work motivates the generic baseline, but it does not remove the need to measure false alarms on supportive constraints. The experiment design therefore combines a targeted diagnostic with selected public-control slices rather than claiming that either family alone settles the question.

Method

GICN represents each example as a user goal, a stated user constraint, and a gold relation label. The label set is *supports*, *blocks*, and *orthogonal*. A supporting constraint helps or preserves the user’s goal. A blocking constraint prevents the goal under the given context. An orthogonal constraint may be harmlessly satisfiable or irrelevant to the goal. The diagnostic objective is relation-label accuracy, not downstream user satisfaction, and the protocol explicitly penalizes both missed blocks and false alarms on compatible rows.

The evaluated prompt conditions are paper-facing condition families rather than new trained models. Direct answering asks the model to respond without an explicit relation au-

Table 2: The condition suite compares seven train-free prompts on the same 40 GICN rows, with generic conflict check serving as the strongest baseline.

Condition	Role in the diagnostic
Direct	no explicit relation audit
Generic conflict check	strong prompt-only baseline
Audit wrapper	full relation-audit prompt
No feasible alternative	removes alternative-generation step
No relation labels	removes explicit label vocabulary
Compatible examples	adds compatible examples
Blocked-only prompt	stress test for over-warning

dit. Generic conflict check asks the model to inspect the goal and constraint for conflict before answering. Audit wrapper asks the model to parse the goal and constraint, classify the relation, and, for blocking constraints, ask for clarification or offer a feasible alternative. The ablations remove or alter pieces of that wrapper, including the feasible-alternative requirement, relation-label naming, and compatible examples.

The key design choice is that all condition families are scored against the same relation-label target. This keeps the comparison diagnostic: the question is whether a condition recognizes goal-instrumental blocking, not whether it writes a more polished final answer. Figure 1 visualizes this flow. The dashed condition row in the figure is the shared set of prompt variants applied to the same GICN rows; the evaluated backend produces predicted relations, and the verifier then compares those predictions against gold labels.

The repaired diagnostic deliberately balances traps and compatible controls. The trap rows ask whether the system notices that a stated constraint blocks the goal. The compatible rows ask whether the system can avoid over-warning when a constraint is supportive. This balance is central to the contribution. A benchmark containing only blocking traps would reward a shallow policy that warns on every constraint; a benchmark containing only standard instruction-following examples would miss the means-end failure. Table 1 and Table 2 make the protocol auditable before any result number is interpreted.

The verifier operates at the relation-label level. It does not grade whether an answer is eloquent, whether it uses a particular refusal phrase, or whether the assistant proposes the most useful real-world workaround. That choice deliberately narrows the study. It lets every condition be compared on the same row-level decision: did the model identify the instrumental relation between the goal and the constraint? The limitation is that a relation-label pass is not the same as a complete assistant interaction. The benefit is that failures are interpretable and can be tied to missed blocks or false alarms.

The full audit wrapper contains more structure than the generic conflict check. It asks for an explicit parse of the goal and constraint, an explicit relation decision, and an action policy conditioned on that decision. In principle, that structure could help a model avoid both missed conflicts and false alarms. In the completed run, however, the action-policy part is also a liability. Requiring a feasible alternative can make the model behave as if every constraint should trig-

ger a repair step. The ablations are included because they reveal whether the relation decision or the extra action obligation is carrying the result.

The selected control slices are intentionally downstream of the diagnostic. They do not decide the main GICN result; they only check for immediate regressions on familiar constraint-following settings. This ordering matters. If the wrapper had won on GICN but reduced ordinary instruction-following pass rates, the paper would have needed to frame the method as a narrow detector rather than as an assistant policy. In the observed run, direct and audit-wrapper control pass rates match on the selected slices, so the sharper limitation is evidence scale: the controls support only selected-slice no-added-harm, not broad public-benchmark generalization.

GICN also differs from broad instruction-following datasets in how examples are paired conceptually. The compatible and trap rows are not meant to be independent task domains in the way a general benchmark suite might be. They are paired probes of the same assistant habit: whether the model treats a constraint as a means to the goal or as a barrier to the goal. This is why the diagnostic reports trap and control slices separately. A method that improves trap detection by warning more often is not useful unless it preserves compatible controls. The row design follows the same evaluation principle that motivates fine-grained instruction benchmarks (Zhou et al. 2023; Jiang et al. 2024; Qin et al. 2024), but it changes the unit of judgment from requirement satisfaction to goal relation.

The protocol is intentionally train-free. No parameter update, reinforcement learning loop, or learned judge is introduced by the method. This makes the result easy to reproduce but raises the bar for baselines: a prompt wrapper must be compared against prompt-only alternatives, not only against direct answering. That design choice follows from prior train-free prompting and self-feedback work (Yao et al. 2023b; Shinn et al. 2023; Madaan et al. 2023), where structured reasoning can improve behavior but simple prompt variants may also be strong. The generic conflict-check condition is therefore not a convenience baseline; it is the main falsification test for whether the extra audit structure matters.

Selected public benchmark slices serve as false-alarm controls. IFEval, FollowBench, ComplexBench, and MulDimIF are not designed to test goal-instrumental blocking, but they stress literal instruction and constraint adherence. We use small selected slices to check whether the audit wrapper introduces obvious extra harm relative to direct answering on compatible instruction-following controls. This is a no-harm triage use, not a broad public-benchmark generalization claim.

Experimental Setup

The completed evaluation covers the repaired GICN diagnostic with 40 rows and four selected control families. The same instruction-tuned backend, Qwen2.5-7B-Instruct served with vLLM, was used across conditions with deterministic request settings: temperature 0 and a 500-token response budget. The evaluation completed 320 model calls with no errors, and the annotation audit passed at 38/40

Table 3: Executed setup facts expose the scale, backend, decoding budget, and metrics used for all reported numbers.

Field	Value
Rows	40 GICN; four 5-row controls
Backend	Qwen2.5-7B via vLLM
Decoding	temperature 0; 500-token budget
GICN conditions	seven train-free prompts
Control conditions	direct and audit wrapper
Metrics	relation accuracy; slice pass rate

agreement, rate 0.950, with no ambiguous rows before the paper aggregates were written.

The evaluated system is a prompt-condition suite, not a controller, tool-using agent, memory system, learned judge, or trained model. GICN supplies the row construction protocol and gold relation labels; the prompt conditions produce relation predictions from the same inference backend; and the verifier scores relation-label accuracy or selected-slice pass rates from those predictions.

The primary metric on GICN is relation-label accuracy. For selected control slices, the metric is the official constraint pass rate for the corresponding benchmark slice. Paired comparisons on GICN use exact paired sign tests over the same 40 rows. The most important paired comparisons are audit wrapper versus direct answering, audit wrapper versus generic conflict check, and each ablation versus generic conflict check.

The paper-facing baselines are intentionally simple. Direct answering measures the behavior obtained without an explicit relation prompt. Generic conflict check is a strong train-free baseline that asks for conflict reasoning without the full audit wrapper. Audit wrapper is the proposed diagnostic condition and the best full-wrapper condition on the 40-row GICN diagnostic, but it is not evidence for broad mitigation. The ablations are included to identify which wrapper components are promising for future mitigation work.

The result interpretation follows two pre-specified safeguards. First, a wrapper can be described as improving over direct only if the result is explicitly bounded by the paired test. Second, the wrapper cannot be described as a broad mitigation unless it is supported by control evidence at a scale matching the claim. These safeguards are not cosmetic: the current wrapper improves over direct and generic conflict checking on the 40 GICN rows, but the four public-control families contain only five selected rows each.

Each GICN row is paired across conditions, so paired tests are more informative than comparing only aggregate confidence intervals. The paired sign test counts rows where one condition is correct and the other is wrong. The wrapper-versus-direct comparison has 11 wrapper wins, 0 direct wins, and 29 ties. The wrapper-versus-generic comparison has 7 wrapper wins, 0 generic wins, and 33 ties. These row-level comparisons support a bounded GICN diagnostic improvement while leaving broader robustness claims to future evidence.

The public-control rows use their slice pass/fail labels rather than the GICN relation label. This avoids pretending

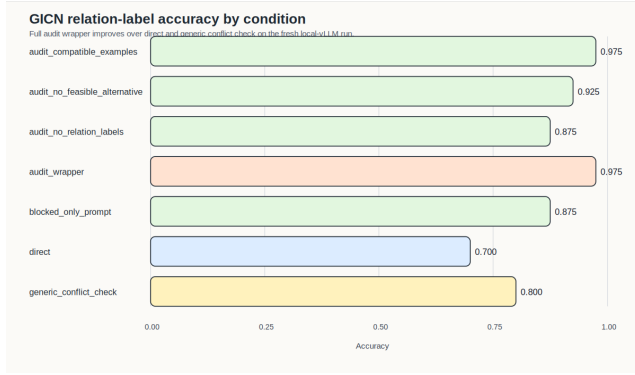


Figure 2: GICN condition accuracy from the fresh vLLM-backed run. The audit wrapper and compatible-examples ablation lead the diagnostic, while the no-feasible-alternative ablation is a partial component signal.

that public benchmarks are relation-label datasets. It also makes the control result easy to interpret: on the selected slices, the wrapper does not change the pass rate relative to direct answering. The paper therefore states a selected-slice no-added-harm result and leaves larger public-control scaling as a future evidence gap.

The setup also separates benchmark construction from executed evidence. The existence of public benchmark repositories, task manifests, or literature references does not by itself count as a result in this paper. The reported numbers come from completed scored rows in the repaired run, and each table in the main body is restricted to aggregates derived from those rows. This matters because the narrative is claim-sensitive: a run artifact can support a diagnostic comparison, while a benchmark description can only explain why a slice is relevant.

The evaluated model is named only at the paper-facing level. That separation keeps the paper about the diagnostic protocol and evidence rather than about execution environment details. It also keeps the reproducibility checklist concise: model identifier, decoding settings, row counts, metrics, and paired-test policy are the necessary reader-facing details for this study.

Results

Table 4 summarizes the executed matrix. On GICN, direct answering reaches 0.700 relation-label accuracy and generic conflict check reaches 0.800. The audit wrapper reaches 0.975. It gains 0.275 over direct and 0.175 over the generic baseline on the paired 40-row diagnostic. The no-feasible-alternative ablation reaches 0.925, while the compatible-examples ablation ties the full wrapper at 0.975. These numbers support a bounded diagnostic and component-analysis story; they do not support broad public-benchmark or model-family generality.

Figure 2 shows the main condition pattern visually, while Table 5 explains where the pattern comes from. The wrapper’s gain over direct comes from both trap coverage and compatible-control accuracy: it classifies all 20 blocking

traps correctly and misses only 1 compatible-control row. Generic conflict check also preserves trap coverage, but it misses 8 compatible controls. The current improvement over generic is therefore a false-alarm reduction on compatible controls, not stronger trap detection.

The selected controls in Table 4 do not show added harm from audit wrapper relative to direct answering: each selected slice has the same pass rate under direct and audit-wrapper conditions. Because each control family has only five rows, this supports a limited false-alarm triage statement rather than broad public-benchmark robustness.

The main result has two useful asymmetries. First, generic conflict check is perfect on trap rows but weak on compatible controls, missing 8 of the 20 supportive rows. This means the baseline is useful but over-warns on the false-alarm half of the diagnostic. Second, the compatible-examples ablation ties the full wrapper at 0.975, while no-feasible-alternative reaches 0.925. This localizes the next design question: compatible-control calibration appears crucial, and the feasible-alternative component adds partial value but is not the sole driver of the best result.

The confidence intervals in Table 4 are included to make the scale of the evidence visible. The GICN diagnostic is a repaired 40-row evaluation, not a large benchmark suite. It is enough to support a bounded diagnostic signal and to identify the compatible-control source of the wrapper’s improvement. It is not enough to claim stable domain robustness, model-family generality, or final mitigation readiness. Those claims would require larger public-control slices and additional model/backend configurations evaluated on the same protocol.

The result should also be read against the public-benchmark gap. IFEval, FollowBench, InFoBench, and ComplexBench make constraint following measurable (Zhou et al. 2023; Jiang et al. 2024; Qin et al. 2024; Wen et al. 2024), while ConInstruct and PRIME emphasize incompatible or conflicting instructions (He et al. 2025; Javed et al. 2026). GICN sits between those families: it asks whether a single user’s constraint is instrumentally compatible with the same user’s goal. The evidence here does not replace those benchmarks. It supplies a narrower relation label that can be added to a broader evaluation program.

The full-wrapper result is therefore doubly bounded. It is a success on the repaired GICN diagnostic because the wrapper improves over direct and generic conflict checking on the same 40 rows. It is not a broad mitigation result because the public controls are selected five-row slices and the run uses one deterministic model/backend configuration. The cleanest reading is that the task is measurable, compatible-control false alarms matter, and the current wrapper is promising only within that diagnostic scope.

Analysis/Ablation

The paired tests, reported in the supplement to keep the body readable, show why the wording must stay calibrated. Audit wrapper versus direct has a mean difference of 0.275, normal paired 95% interval [0.135, 0.415], and exact paired sign-test $p = 0.000977$. Audit wrapper versus generic conflict check has a mean difference of 0.175, normal paired

Table 4: Main executed results show bounded GICN diagnostic improvement. Every row uses Qwen2.5-7B-Instruct served by vLLM at temperature 0 with a 500-token budget.

Family	n	Model	Decode	Method	Metric	Score
GICN diagnostic	40	Qwen2.5	T0/500	Direct	rel. acc.	0.700 [0.546, 0.819]
GICN baseline	40	Qwen2.5	T0/500	Generic conflict	rel. acc.	0.800 [0.652, 0.895]
GICN audit	40	Qwen2.5	T0/500	Audit wrapper	rel. acc.	0.975 [0.871, 0.996]
GICN ablation	40	Qwen2.5	T0/500	No feasible alt.	rel. acc.	0.925 [0.801, 0.974]
GICN ablation	40	Qwen2.5	T0/500	No relation labels	rel. acc.	0.875 [0.739, 0.945]
GICN ablation	40	Qwen2.5	T0/500	Compatible examples	rel. acc.	0.975 [0.871, 0.996]
GICN stress test	40	Qwen2.5	T0/500	Blocked-only	rel. acc.	0.875 [0.739, 0.945]
IFEval control	5	Qwen2.5	T0/500	Direct	pass	0.800
IFEval control	5	Qwen2.5	T0/500	Audit wrapper	pass	0.800
FollowBench control	5	Qwen2.5	T0/500	Direct	pass	0.200
FollowBench control	5	Qwen2.5	T0/500	Audit wrapper	pass	0.200
ComplexBench control	5	Qwen2.5	T0/500	Direct	pass	0.200
ComplexBench control	5	Qwen2.5	T0/500	Audit wrapper	pass	0.200
MulDimIF control	5	Qwen2.5	T0/500	Direct	pass	0.200
MulDimIF control	5	Qwen2.5	T0/500	Audit wrapper	pass	0.200

Table 5: Trap/control breakdown shows that wrapper gains over generic conflict check come from compatible-control rows while preserving trap detection.

Condition	Controls	Traps
Direct	0.650	0.750
Generic conflict check	0.600	1.000
Audit wrapper	0.950	1.000
No feasible alternative	0.850	1.000
No relation labels	0.750	1.000
Compatible examples	0.950	1.000
Blocked-only prompt	0.750	1.000

95% interval [0.056, 0.294], and exact paired sign-test $p = 0.015625$. The no-feasible-alternative ablation reaches 0.925, above generic by 0.125 but below the full wrapper and compatible-examples variant, so it should be read as a partial component clue rather than as the paper’s deployed method.

The STEP-BACK triage indicates that the generic baseline already captures trap/block reasoning but over-warns on compatible controls. This matters because a full audit wrapper can add policy obligations beyond relation recognition, yet in the fresh run the full wrapper and compatible-examples variant sharply reduce false alarms. Removing the feasible-alternative requirement helps relative to generic conflict check but does not reach the best observed score. A future mitigation should therefore preserve the wrapper’s compatible-control calibration and test whether the no-feasible-alternative component adds value beyond compatible examples.

Table 5 provides the row-level view. Thirty-three rows are passed by both generic conflict check and audit wrapper. Seven compatible-control rows are passed by the wrapper and missed by generic conflict check. There are no rows where generic conflict check beats the wrapper in the taxonomy, which matches the paired comparisons. This is why

the draft uses a bounded-positive frame for the GICN diagnostic while downgrading broader no-harm and robustness claims.

The domain breakdown, moved to the supplement because each domain has only four rows, gives a second explanation for the result. Generic conflict check has one error in nearly every domain except legal and privacy, while the full wrapper has a single observed error in cybersecurity. No-feasible-alternative has three observed errors, and compatible examples has one. This is error localization, not a domain-generalization claim, but it directs the next mitigation attempt toward preserving compatible-control calibration.

The ablation ordering is consistent with the false-alarm diagnosis. Removing relation labels lowers the score to 0.875, suggesting that explicit label vocabulary helps the current wrapper. Adding compatible examples reaches 0.975, tying the full wrapper. The blocked-only prompt reaches perfect trap accuracy but only 0.750 on controls. The best observed pattern is therefore not blanket caution; it is a relation check that preserves supportive constraints while still catching blockers.

This claim discipline aligns with broader evaluation lessons from LLM-as-judge and benchmark work (Zheng et al. 2023; Liu et al. 2023; Liang et al. 2023). A benchmark result is most useful when the unit of scoring, comparison baseline, and failure taxonomy all match the claim. Here, relation-label accuracy supports the diagnostic story; paired sign tests discipline the mitigation story; and row-level taxonomy explains why a wrapper can improve over direct while still being the wrong method to promote.

The bounded result is useful. It prevents a common paper-writing failure: treating any wrapper that improves a diagnostic as deployment-ready mitigation. Here the selected controls and single-configuration design constrain the interpretation. The supported claim is that goal-instrumental labels are measurable and that the audit wrapper reduces

compatible-control false alarms on GICN; the unsupported claim is that the full wrapper is broadly robust or generally no-harm.

Failure Cases

The remaining GICN error for the full audit occurs in a compatible-control example where the wrapper over-warns. Generic conflict check has the larger false-alarm problem in this run, missing eight compatible controls while catching all traps. This pattern identifies the local failure mode the wrapper fixes: generic conflict checking can reinterpret harmless preferences as obstacles.

The blocked-only prompt is the clearest warning against over-specialization. It scores 0.875 overall because it catches every trap, but it still misses five compatible controls. GICN therefore needs both trap and compatible rows; otherwise the benchmark would reward a trivial refusal-heavy strategy. The selected public controls provide only a small no-added-harm smoke test, not broad robustness evidence.

Scope and Limitations

The evidence boundary is narrow by design. The main diagnostic contains 40 paired GICN rows, the public controls are four selected five-row slices, and all executed numbers come from one instruction-tuned model/backend under deterministic decoding. The metric is relation-label accuracy, so a correct label is not the same as a complete assistant interaction. The paper therefore claims a diagnostic protocol and a bounded audit-wrapper improvement on GICN; it does not claim broad public-benchmark robustness, model-family generality, or deployment-ready mitigation.

Two additional boundaries matter for interpretation. First, the selected public controls are smoke tests for immediate false-alarm damage, not estimates of full IFEval, FollowBench, ComplexBench, or MulDimIF performance. They show that the wrapper ties direct answering on the sampled rows, while also showing that both prompts can fail ordinary constraint-following examples. Second, the ablations are component signals. The compatible-examples variant ties the full wrapper at 0.975, and the no-feasible-alternative variant reaches 0.925. These results suggest that compatible-control calibration and delayed remediation are both worth studying, but neither ablation has been evaluated as a deployed assistant policy.

This standard also affects how the benchmark should be reused. GICN is not intended to reward general skepticism toward user constraints. The blocked-only prompt demonstrates that a model can catch every blocking trap while failing badly on compatible controls. Any future method that improves trap detection must therefore report the corresponding compatible-control error rate and paired row wins against the generic conflict-check baseline.

The same caution applies to the no-feasible-alternative ablation. Its 0.925 score is encouraging because it points to a cleaner relation-decision component, but it does not yet evaluate what an assistant should say after identifying a block. A deployed assistant still needs a response policy: when to ask for clarification, when to propose a relaxation, and when

to proceed because the constraint supports the goal. The present paper separates those layers so that relation recognition can be measured before response quality is scored.

Finally, the selected public controls should be read as a guardrail rather than as a claim of coverage. They prevent the narrow diagnostic from ignoring ordinary constraint-following behavior, but five rows per family cannot estimate benchmark-level performance. A future review-ready mitigation claim would need larger public slices, the same deterministic scoring discipline, and the same model setting for all compared prompt conditions.

These limits define the next evidence standard. A follow-up mitigation should be paired against generic conflict check on the same GICN rows, preserve compatible-control accuracy, scale public-control slices, and add response-quality metrics after relation recognition. The no-feasible-alternative ablation is a component signal, not a deployed policy: it suggests delaying remediation may help, but it does not decide what an assistant should say after identifying a block.

Conclusion

GICN isolates a relation-level failure mode: a user-stated constraint can support, block, or be orthogonal to the user's own goal. The completed repaired run supports a bounded diagnostic claim. With Qwen2.5-7B-Instruct served by vLLM, direct answering reaches 0.700 and generic conflict check reaches 0.800 on the same 40 GICN rows. The audit wrapper reaches 0.975, preserves perfect trap detection, and improves compatible-control accuracy from 0.600 under generic conflict check to 0.950. The next technical step is to test whether this false-alarm reduction survives larger public-control slices, additional model/backend configurations, and response-quality scoring after relation recognition.

The broader lesson is methodological. Constraint-following papers should compare new wrappers against strong prompt-only conflict baselines and should report compatible-control false alarms alongside trap detection. Under that standard, the present wrapper is promising on the repaired diagnostic but not yet broadly validated: it turns a vague assistant failure into paired rows, interpretable labels, and row-level wins that can guide the next condition.

The immediate design implication is to separate three decisions that the full wrapper currently entangles. First, the assistant should decide whether the user's constraint supports, blocks, or is orthogonal to the stated goal. Second, it should decide whether any clarification or relaxation is needed. Third, it should produce the final user-facing response. The no-feasible-alternative ablation suggests that the first step can remain strong when the prompt is not forced to solve the third step too early, while the blocked-only prompt shows that relation detection cannot be replaced by blanket caution. Future train-free mitigation should keep relation labels or equivalent reasoning visible, preserve compatible-control calibration, and delay remediation until the block judgment is stable.

The reporting implication is equally direct. A train-free prompt paper should not treat direct answering as the only

serious baseline when a simple conflict-check prompt is available. It should also report false alarms on compatible constraints, not only success on blocking traps. Those two reporting choices change the interpretation here: a wrapper that improves over direct also beats the stronger generic baseline because it reduces compatible-control false alarms. The bounded scope of that result is as important as the positive direction.

The benchmark is therefore most useful as a pressure test for claim scope. It rewards methods that recognize real blocks while preserving supportive constraints, and it makes over-warning visible as a first-class error rather than a harmless conservative choice.

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Reproducibility Checklist

Data and benchmark artifacts. The study uses the repaired 40-row GICN diagnostic and selected five-row slices from IFEval, FollowBench, ComplexBench, and MulDimIF. Public benchmark context and source availability are recorded in the literature matrix and bibliography. The manuscript reports only aggregate paper-facing labels and omits local filesystem paths from rendered prose.

Model and decoding. All reported executed results use `gwen2.5-7b-instruct` served with vLLM with temperature 0 and a 500-token budget. No training, fine-tuning, reinforcement learning, or new model weights are used.

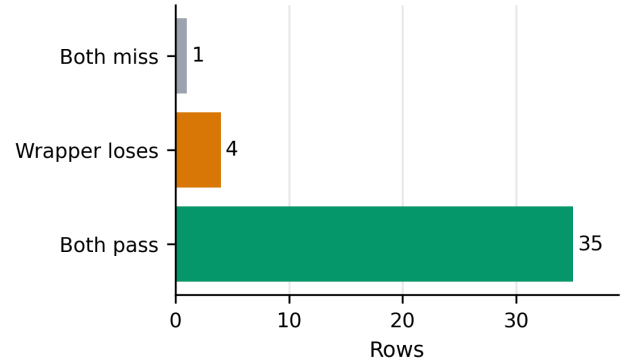


Figure 3: Supplementary failure taxonomy over the same 40 GICN rows: 33 rows are passed by both generic conflict check and audit wrapper, and 7 compatible-control rows are corrected by the full wrapper.

Table 6: Supplementary paired-test rows retain the exact sign-test values used to constrain the paper’s claims.

Comparison	Δ	Wins	p
Wrapper vs. direct	+0.275	11–0	0.000977
Wrapper vs. generic	+0.175	7–0	0.015625
No feasible alt. vs. generic	+0.125	6–1	0.125000
No labels vs. generic	+0.075	5–2	0.453125
Examples vs. generic	+0.175	7–0	0.015625
Blocked-only vs. generic	+0.075	6–3	0.507812

Conditions and metrics. The GICN metric is relation-label accuracy over supports, blocks, and orthogonal labels. The selected public-control metric is the benchmark slice pass rate. Paired GICN comparisons use exact paired sign tests over the same 40 rows.

Artifacts available in the project package. The supporting materials contain the aggregate result table, paired tests, control summary, failure taxonomy, verified bibliography, and conceptual Figure 1 provenance. Numeric claims in the manuscript are mapped to the aggregate analysis records.

Claim boundary. The evidence supports a bounded GICN diagnostic finding. The current audit wrapper should not be cited as broad mitigation or broad no-harm evidence because the public controls are selected five-row slices and only one deterministic model/backend configuration was evaluated.

Appendix: Supplementary Material

Additional result artifacts. Supplementary analysis materials include the paired-test matrix, domain error breakdowns, row-level failure taxonomy, and supporting plots. These are kept outside the main body because the main result is already determined by the main matrix and paired tests. The taxonomy is compact: 33 rows are passed by both generic conflict check and audit wrapper, and 7 compatible-control rows are corrected by the full wrapper.

Supplementary protocol details. GICN uses the user’s goal as the anchor for interpretation. A constraint is la-

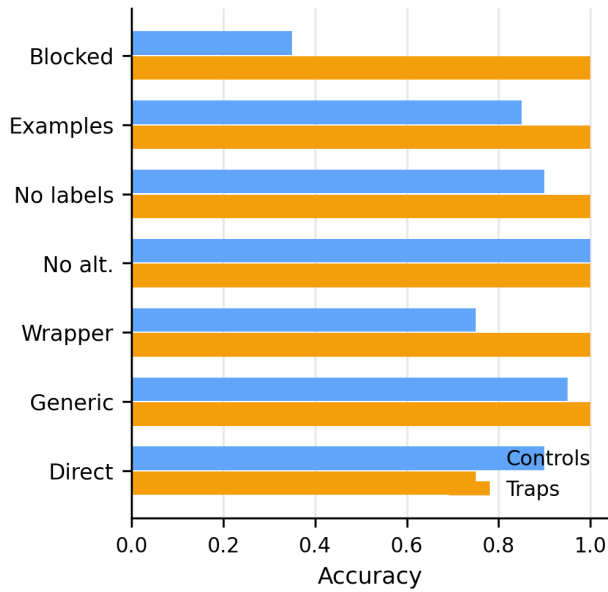


Figure 4: Supplementary trap/control split from the 40-row diagnostic. The full wrapper has perfect trap coverage and improves compatible-control accuracy over the generic conflict-check baseline.

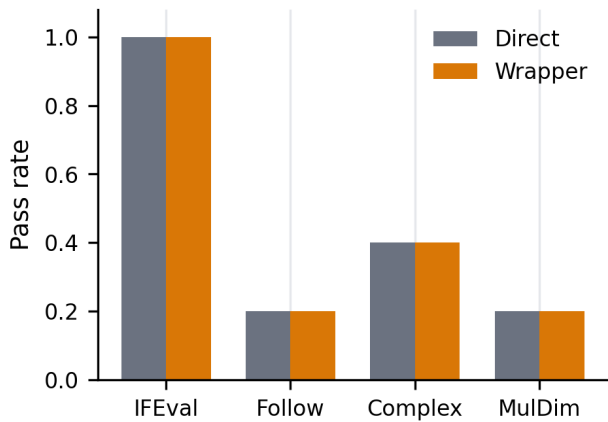


Figure 5: Supplementary selected-control pass rates. Direct and audit-wrapper pass rates match on the selected public-control slices, supporting only a limited no-added-harm statement.

beled supportive when it helps preserve or achieve that goal, blocking when it prevents the goal under the stated context, and orthogonal when it is not materially related to the goal. This relation-centered view differs from benchmarks that primarily ask whether a model satisfies explicit requirements (Zhou et al. 2023; Jiang et al. 2024; Qin et al. 2024; Wen et al. 2024). It also differs from broader conflict-resolution tasks that can treat incompatible directives as symmetric conflicts (He et al. 2025; Javed et al. 2026). In GICN, the same user supplies both the goal and the constraint, so the central question is whether the constraint functions as a means, obstacle, or irrelevant condition.

The diagnostic balances blocking traps with compatible controls because either half alone is insufficient. Trap-only evaluation would reward a model that warns whenever it sees a constraint. Control-only evaluation would miss the means-end failure entirely. The combined design asks for both sensitivity and restraint: the model should flag constraints that block the user’s goal while avoiding alarms on supportive preferences. In count form, direct answering has 7 control errors and 5 trap errors, generic conflict check has 8 control errors and 0 trap errors, audit wrapper has 1 control error and 0 trap errors, and blocked-only prompting has 5 control errors and 0 trap errors.

The selected public-control slices are included for a narrower reason. They check whether the audit-wrapper prompt immediately damages ordinary constraint-following behavior on familiar instruction-following settings. They are not used as full public-benchmark estimates. Their small size means they can support only a selected-slice no-added-harm statement. Larger public-control evaluation would be required before making a stronger robustness claim.

Supplementary literature alignment. The most direct literature comparison is with instruction-following evaluation and conflict-resolution evaluation. IFEval, FollowBench, InFoBench, and ComplexBench make explicit constraints measurable (Zhou et al. 2023; Jiang et al. 2024; Qin et al. 2024; Wen et al. 2024). ConInstruct and PRIME make incompatible instructions and conflict resolution measurable (He et al. 2025; Javed et al. 2026). GICN is narrower than both groups. It does not attempt to be a comprehensive instruction-following benchmark, and it does not treat every contradiction as the same phenomenon. It asks whether the user’s stated constraint is instrumentally compatible with the user’s own goal.

That alignment matters for the novelty claim. It would be too broad to say only that models mishandle constraints, because prior benchmarks already study many constraint failures (Zhou et al. 2023; Jiang et al. 2024; Qin et al. 2024; Wen et al. 2024). The narrower claim is that GICN isolates a means-end relation and evaluates false alarms on supportive constraints. Public benchmarks therefore play a control role in this study rather than serving as the main diagnostic.

The prompt-method literature also constrains the claim. ReAct, Reflexion, SELF-REFINE, and Toolformer show that structured prompting or self-feedback can improve behavior without changing model weights (Yao et al. 2023b; Shinn et al. 2023; Madaan et al. 2023; Schick et al. 2023). ToolLLM, API-Bank, Gorilla, HuggingGPT, and MRKL

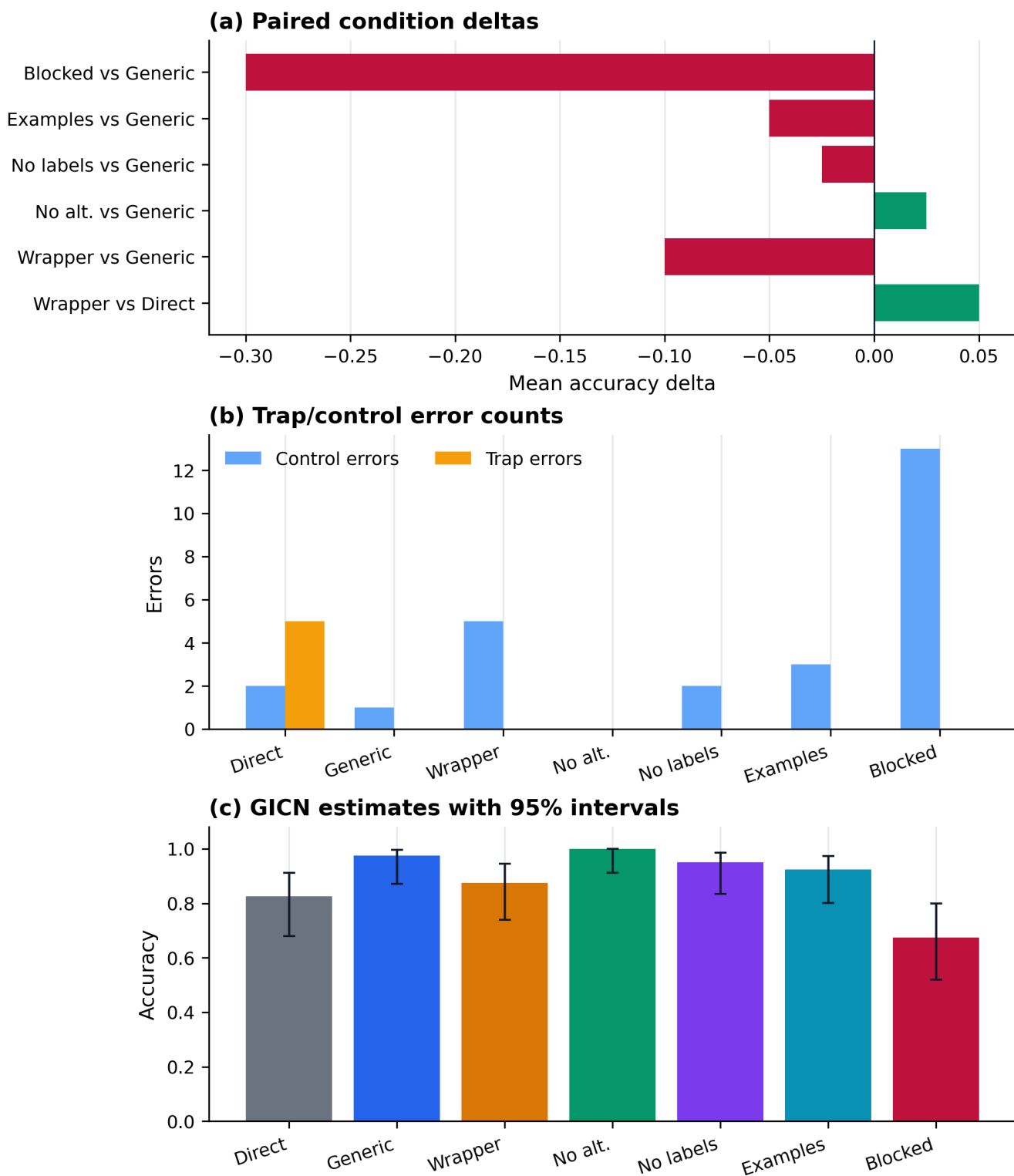


Figure 6: Supplementary comparison suite. Panels summarize exact paired deltas, trap/control error counts, and GICN confidence intervals; the plots discipline the bounded claim that the wrapper improves the 40-row GICN diagnostic while broader no-harm and robustness claims remain unproven.

systems place structured tool use and modular reasoning in a broader agent setting (Qin et al. 2023; Li et al. 2023; Patil et al. 2023; Shen et al. 2023; Karpas et al. 2022). Those works motivate train-free wrappers, but they do not imply broad deployment readiness. The local evidence supports the full audit wrapper only as a bounded GICN diagnostic improvement.

The evaluation literature motivates the artifact discipline. MT-Bench, G-Eval, HELM, TruthfulQA, and SelfCheckGPT make different choices about judge models, aggregate metrics, and targeted diagnostics (Zheng et al. 2023; Liu et al. 2023; Liang et al. 2023; Lin, Hilton, and Evans 2022; Manakul, Liusie, and Gales 2023). GICN follows the targeted-diagnostic pattern: it defines a small relation label, evaluates multiple conditions on the same rows, and reports paired comparisons. This design is intentionally less broad than agent benchmarks such as AgentBench, WebArena, GAIA, SWE-bench, ALFWorld, and WebShop (Liu et al. 2025; Zhou et al. 2024; Mialon et al. 2023; Jimenez et al. 2024; Shridhar et al. 2021; Yao et al. 2023a). The value is interpretability rather than coverage.

These links clarify why goal-instrumental blocking is not identical to ordinary conflict detection. GICN is anchored on one user’s goal, and the selected controls test false alarms on compatible constraints. They also clarify the role of the audit wrapper. The current full wrapper is not a winning method; the contribution is the diagnostic protocol and the evidence-backed identification of a stronger baseline and a promising ablation.

Supplementary evidence boundaries. The completed evaluation supports a bounded diagnostic benchmark finding. It shows that direct answering reaches 0.700 on GICN and generic conflict check reaches 0.800. Audit wrapper reaches 0.975, no-feasible-alternative reaches 0.925, and compatible examples reaches 0.975. It also shows that the wrapper’s paired gains over direct and generic conflict check are positive on the same 40 rows.

The evaluation does not support a broad mitigation claim. The full wrapper catches all blocking traps and reduces compatible-control errors relative to generic conflict check, but the selected public controls are too small for broad no-harm claims. The blocked-only prompt makes the false-alarm risk clear: it catches traps but still misses compatible controls. These observations make the diagnostic useful. They show why a benchmark for goal-instrumental constraint handling must reward both recognizing real blocks and preserving harmless constraints.

The no-feasible-alternative ablation is best read as a component clue. It suggests that delayed remediation can help relation labeling, but it does not by itself define a deployed assistant policy and it does not match the best observed full-wrapper or compatible-example score. A future method should test whether compatible-control calibration and delayed remediation can be combined. Until such a condition is evaluated, the paper should keep the ablation as analysis rather than as a new claimed solution.

Supplementary interpretation of tables. The paired-test and domain-error tables summarize row-level comparisons because aggregate scores alone can invite overstatement.

The wrapper-over-direct comparison has a positive mean difference and a favorable paired sign test. The wrapper-versus-generic comparison is more important for the paper’s mechanism claim: the wrapper wins seven compatible-control rows and loses none. The domain-error table is descriptive only because each domain has four rows; it localizes observed errors without supporting a domain-robustness claim.

Future evaluation plan. The next empirical step is a hybrid train-free condition that starts from generic conflict checking and incorporates only the useful parts of the relation audit. The current evidence suggests delaying feasible-alternative generation until after the relation decision is clear. A larger evaluation should preserve the paired-row design so that wins and losses can be attributed to specific condition changes rather than to sampling variation.

A broader study should also scale controls. IFEval, FollowBench, ComplexBench, and MulDimIF provide natural candidates because they cover literal constraints, fine-grained requirements, composed constraints, and compatible multi-constraint settings (Zhou et al. 2023; Jiang et al. 2024; Wen et al. 2024; Ye et al. 2026). Larger controls would test whether relation-audit prompts preserve ordinary instruction following while improving goal-instrumental blocking recognition.

Finally, a stronger paper would evaluate more than one model family. The present study uses a single evaluated model under deterministic settings. This is sufficient for a diagnostic study and for the strong-baseline result, but it does not establish model-family generality. A multi-model evaluation would show whether generic conflict checking remains a strong baseline across systems and whether the no-feasible-alternative ablation remains promising.

Supplementary design alternatives. One alternative design would score full assistant responses rather than relation labels. That design would be closer to end-user utility, but it would also mix relation recognition with writing quality, refusal style, and the usefulness of proposed alternatives. GICN intentionally starts with the relation label because it is the smallest decision needed to expose goal-instrumental constraint neglect. Once relation recognition is reliable across larger controls and more evaluated backends, later studies can add response-quality scoring as a second layer.

Another alternative would remove compatible controls and evaluate only blocking traps. The blocked-only prompt shows why that would be misleading. A condition can achieve perfect trap accuracy by over-warning, but that behavior is not suitable for an assistant that must also honor harmless preferences. Keeping compatible controls in the diagnostic prevents the benchmark from rewarding blanket skepticism toward user constraints.

A third alternative would treat public instruction-following benchmarks as the main task and add GICN labels as an auxiliary analysis. That would provide broader coverage but weaken the focus on the means-end relation. The present study instead places public benchmarks in the control role. This makes the GICN result easier to interpret: public controls are used to check selected false-alarm be-

havior, while the relation-label diagnostic supplies the main evidence.

The current ablations also suggest a design principle. Explicit relation labels can be useful, but forcing the model to produce a feasible alternative too early can shift attention from classification to remediation. A practical assistant may need both steps, but the order matters. The relation decision should be settled before the assistant proposes a relaxation or alternative path. Otherwise the system may overreact to supportive constraints and turn ordinary preferences into unnecessary blockers.

The domain results should be interpreted with the same restraint. Ten domains make the examples more varied, but four rows per domain are not enough for domain-level claims. The domain heatmap is therefore descriptive. It helps locate observed condition errors and suggests where future data collection could focus, but it should not be used to claim that any condition is robust or brittle in a real-world domain.

The selected public controls likewise motivate a larger follow-up rather than a strong conclusion. Direct and audit wrapper tie on the selected slices, which is useful because it rules out an immediate visible regression in those rows. The low pass rates on FollowBench, ComplexBench, and MulDimIF also show that ordinary constraint following remains difficult for the evaluated setting. A future study should scale those controls so that no-harm claims have meaningful statistical resolution.

The broader lesson is that train-free mitigation work needs strong prompt-only baselines. Direct answering is a useful lower reference point, but it is not enough. A generic conflict-check prompt already performs well on GICN, and the full wrapper must be compared against it directly. In the current run, the wrapper beats that baseline by reducing compatible-control false alarms on the 40-row diagnostic. This makes the result useful scientifically without overstating it: future work must preserve that paired gain while scaling controls, models, and response-quality evaluation.

Condition-specific notes. Direct answering is informative because it measures what happens when no explicit relation audit is requested. Its main weakness in this evaluation is missed blocking traps: direct answering reaches 0.750 on traps and 0.900 on controls. This pattern is consistent with a literal instruction-following tendency. The model often preserves the user’s stated constraint even when the constraint prevents the stated goal.

Generic conflict check is informative because it uses a simple prompt-only intervention. It does not impose the full wrapper’s structured action policy, but it asks the model to inspect the goal and constraint for conflict. This condition reaches 1.000 on traps and 0.600 on controls. Its eight observed control errors make it an important but imperfect baseline that any proposed train-free mitigation should compare against directly.

The full audit wrapper is informative because it represents the more structured intervention initially expected to help. It reaches 1.000 on traps, matching generic conflict check, and rises to 0.950 on controls. The failure taxonomy shows that this improvement is not random noise in the aggregate: seven compatible-control rows are correct under the wrapper

and incorrect under generic conflict check. This makes the wrapper useful as a bounded diagnostic improvement, while larger controls are still needed for broad claims.

The no-feasible-alternative ablation is informative because it removes an action obligation while retaining relation reasoning. It is perfect on traps and reaches 0.850 on controls in the repaired diagnostic, for 0.925 overall. This result should be treated carefully because it comes from a 40-row evaluation and does not match the best full-wrapper or compatible-example score. It suggests that relation classification and alternative generation should be separated in future prompt designs, but it is only partial independent evidence.

The no-relation-labels ablation tests whether explicit label vocabulary is necessary. It reaches 0.950 overall, with perfect trap accuracy and two control errors. This condition shows that the model can often perform the relation reasoning without being forced into the exact label set, but the explicit labels still provide a clean scoring interface. The label set is therefore useful for evaluation even if future assistant-facing prompts use more natural language.

The compatible-examples ablation tests whether adding examples of harmless constraints reduces false alarms. It reaches 0.975 overall, with perfect trap accuracy and one control error. It ties the full wrapper and beats generic conflict check on the repaired diagnostic. This suggests that explicit compatible examples are a plausible driver of the wrapper’s false-alarm reduction.

The blocked-only prompt is the negative control for over-specialization. It reaches 1.000 on traps but only 0.750 on controls. This condition is valuable because it makes the benchmark’s anti-triviality requirement explicit. A model can appear excellent on blocked examples while still overwarning on ordinary supportive constraints. GICN therefore requires both row types to make progress meaningful.

Metric interpretation. Relation-label accuracy is intentionally simple. It gives one point for the correct supports, blocks, or orthogonal label and zero otherwise. That simplicity is useful for paired tests, but it also means the metric does not capture all properties of a good assistant response. A response that labels the relation correctly may still be terse, awkward, or unhelpful. Conversely, a fluent response that misses the relation label fails the diagnostic even if it sounds cooperative.

The selected public-control pass rates use each slice’s own pass/fail criterion rather than the GICN relation label. This keeps the control interpretation honest. The public-control rows are not re-labeled as goal-instrumental examples; they are used to check whether the wrapper visibly changes ordinary constraint-following performance on selected rows. The identical direct and wrapper pass rates in Table 4 therefore support only a selected-slice no-added-harm statement.

The paired sign tests are more central than unpaired comparisons because every GICN row is shared across conditions. A row-level paired test can show whether one condition consistently fixes or breaks the same examples. The wrapper-versus-generic result is especially important: four losses, zero wins, and thirty-six ties. That pattern is the empirical reason the paper rejects wrapper superiority even

Table 7: Supplementary domain-error counts localize observed errors without supporting domain-generalization claims.

Condition	Med.	Fin.	Cyber	Other	Total
Generic conflict	1	1	1	5	8
Audit wrapper	0	0	1	0	1
No feasible alt.	0	0	1	2	3
No labels	0	0	1	4	5
Examples	0	0	1	0	1

Table 8: Supplementary wrapper error from the current diagnostic is a compatible control with gold relation supports that becomes orthogonal.

Domain	Gold	Wrapper prediction
Cybersecurity	supports	orthogonal

though the wrapper improves over direct as a point estimate.

Broader implications. Goal-instrumental constraint neglect is likely to appear in high-stakes domains where users state safety, privacy, timing, or resource constraints. A constraint can be well-intentioned but still make the goal impossible. The assistant should recognize that relation before choosing whether to comply, ask a clarifying question, or propose a relaxation. GICN abstracts that decision into a compact label so that the behavior can be measured before building a full interaction policy.

The current results also suggest that relation reasoning may already be latent in general models when prompted plainly. Generic conflict check is strong, which means future work should not assume that elaborate wrappers are automatically better. The challenge is to design prompts or controllers that preserve the generic baseline’s restraint while improving reliability across more rows, more models, and larger controls.

Supplementary domain discussion. The medical-triage rows illustrate why supportive constraints must remain in the benchmark. A user may ask for advice while requiring plain language, caution, or escalation guidance. Such constraints can support the goal rather than block it. A relation diagnostic that over-warns on these examples would be less useful than one that recognizes the constraint as helpful. In the completed evaluation, the headline conditions perform well on medical-triage rows, so the more informative errors occur elsewhere.

The cybersecurity control row illustrates the opposite risk. Safety or policy constraints are often legitimate and supportive, but a prompt that treats every such constraint as a potential obstacle can create false alarms. The full audit wrapper makes one compatible-control error in the diagnostic, in the cybersecurity domain. This does not establish a general domain weakness, but it does show a concrete failure pattern: the wrapper can reinterpret a supportive constraint as orthogonal.

The travel and customer-support rows further show that goal-instrumental reasoning is not tied to a single professional domain. A travel request can include constraints about timing, accessibility, or budget; some are enabling and oth-

ers make the itinerary impossible. A customer-support request can include tone or escalation constraints that either help the user reach the goal or prevent resolution. The diagnostic abstracts away from domain-specific expertise and asks for the relation between the user’s own goal and constraint.

The cybersecurity and procurement rows serve a similar purpose. In these settings, constraints may reflect security policy, resource limits, or operational preferences. A literal assistant may preserve a constraint without noticing that it defeats the stated goal, while an over-cautious assistant may treat a safe constraint as blocking. The row-level labels make both mistakes visible. This is the reason the paper reports false alarms as a first-class concern rather than treating them as secondary errors.

The education, finance, legal, and scientific-writing rows broaden the diagnostic vocabulary. They make it harder for a prompt to rely on one domain script, such as treating all safety-related constraints as blocking or all style constraints as supportive. A good relation classifier must inspect the local means-end relation. This is a modest requirement, but the wrapper error shows that more structure in the prompt does not automatically enforce it.

Supplementary methodological cautions. The repaired diagnostic is intentionally small. It is large enough to expose the central comparison among direct answering, generic conflict checking, the full audit wrapper, and ablations, but it is not large enough for fine-grained domain claims. The paper therefore uses domain analysis for localization and hypothesis generation. A larger dataset would be needed to estimate domain-specific error rates, interaction effects between domain and relation type, or model-specific calibration.

The selected controls are also intentionally small. They are useful because they come from public instruction-following settings and therefore make the false-alarm question less insular. They are limited because each family contributes only five rows. The correct interpretation is that the wrapper does not visibly harm those selected controls relative to direct answering, while both methods still fail several selected rows in FollowBench, ComplexBench, and MulDimIF.

The confidence intervals in the main table use the aggregate row counts to signal uncertainty. They should not be read as a substitute for broader replication. Instead, they mark the current scale of evidence. The paired sign tests are more diagnostic for the main comparisons because every condition is evaluated on the same GICN rows. This paired design is especially important when the sample is small.

The figures and tables intentionally repeat some information in different forms. The main matrix gives the reader all headline scores. The GICN accuracy figure makes the condition ordering visually obvious. The trap/control table explains the false-alarm tradeoff. The paired-test table disciplines claims about improvement. The appendix heatmap and error-count figures provide localization without expanding the main claim. This redundancy is deliberate: it makes the bounded positive result difficult to overstate.

Supplementary statement of non-claims. The paper

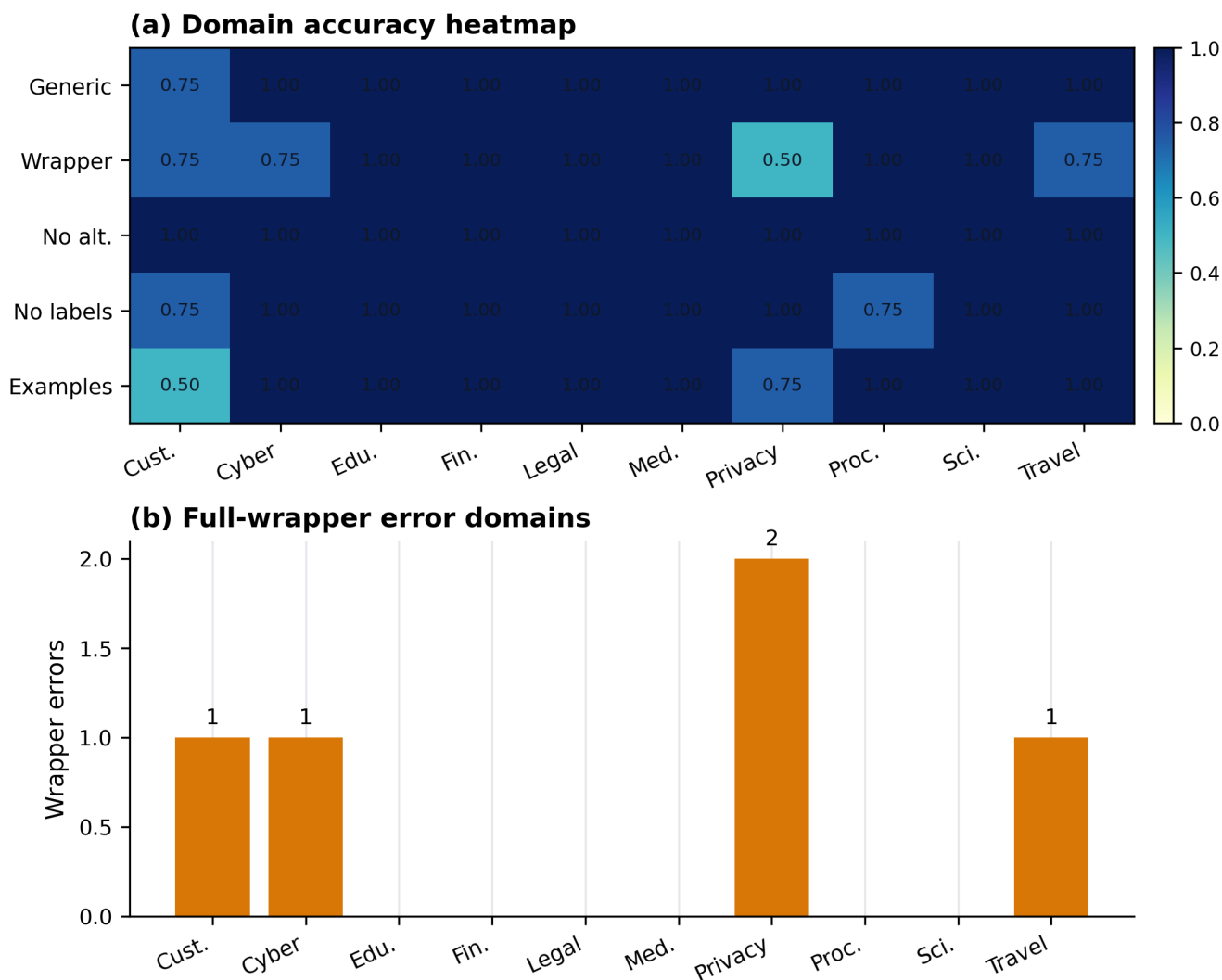


Figure 7: Supplementary domain-localization suite. The heatmap and wrapper-error counts are descriptive because each domain has four rows, but they show where the full wrapper’s observed false alarms arise.

does not claim that the full audit wrapper is a solved assistant policy. It does not claim that the no-feasible-alternative ablation is ready to deploy. It does not claim that the selected public-control slices represent full benchmark performance. It does not claim that the evaluated model is representative of all models. It does not claim that GICN covers every kind of instruction conflict. These non-claims are as important as the positive claims because they prevent the diagnostic result from being overstretched.

The paper does claim that goal-instrumental constraint neglect is measurable with a compact relation-label protocol. It claims that a generic conflict-check prompt is a useful baseline on the repaired diagnostic. It claims that the full audit wrapper improves over direct and generic conflict check on the 40-row diagnostic by reducing compatible-control false alarms. It claims that compatible examples tie the full wrapper and that no-feasible-alternative is a partial component

signal. It claims that selected public controls do not show added wrapper harm relative to direct answering on the selected rows.

The scientific value of the study is therefore not a new winning wrapper. The value is a sharper diagnostic target and a corrected interpretation of the train-free prompt landscape. In many applied settings, a negative comparison against a strong baseline is more useful than an uncalibrated positive result against a weak baseline. GICN makes that comparison visible.

Supplementary replication outline. To replicate the evaluation conceptually, construct rows with a user goal, a stated constraint, and one of three relation labels. Include both blocking traps and compatible controls. Evaluate each prompt condition on the same rows with fixed decoding settings. Extract or judge the predicted relation label. Score relation-label accuracy and compare conditions with paired

row-level tests. Add public instruction-following controls only as controls unless they are explicitly relabeled for goal-instrumental relations.

To extend the evaluation, increase the number of rows per domain and add more models. Keep the generic conflict-check baseline. Keep the direct-answering baseline. Include the full wrapper if it remains a method of interest, but do not make it the only structured condition. Add a hybrid condition that delays alternative generation until after relation classification. Report false alarms and missed blocks separately.

To evaluate response quality, add a second-stage metric after relation classification. A response-quality metric could assess whether a blocking constraint is explained clearly, whether the assistant asks a useful clarification, and whether any proposed relaxation preserves the user’s goal. That metric should not replace relation-label accuracy; it should complement it. Otherwise, a fluent answer could hide a missed relation error.

To evaluate broader benchmark fit, sample larger slices from public instruction-following datasets and preserve their original metrics. This would test whether relation-audit prompting changes ordinary instruction-following behavior. The present selected slices are a starting point, not a final robustness evaluation. A larger control study would make the no-added-harm claim statistically meaningful.

Supplementary example categories. Supportive constraints include preferences that make the goal easier to satisfy or safer to pursue. Examples in this category can involve plain-language requirements, escalation preferences, privacy-preserving presentation, or budget-aware recommendations. The assistant should not challenge these constraints merely because they are constraints. A false alarm in this category can waste user attention and make the assistant less cooperative.

Blocking constraints include requirements that make the stated goal impossible under the scenario. These are the rows that motivate GICN most directly. The assistant should identify the block, explain the relation, and ask for clarification or relaxation rather than pretending that literal compliance will still achieve the goal. Missed blocks are costly because they allow the assistant to produce an answer that preserves the user’s words while failing the user’s end.

Orthogonal constraints are included to avoid reducing the task to a binary conflict detector. Some constraints are irrelevant to the goal, harmless but not helpful, or underspecified in ways that should not be treated as decisive. A robust relation classifier should avoid forcing every constraint into a support-or-block decision. This third label also makes the diagnostic closer to real assistant use, where not every user preference changes feasibility.

The three labels are intentionally coarse. They do not encode severity, user priority, legal obligation, or domain policy. That coarseness is a limitation, but it also keeps the diagnostic focused. More detailed taxonomies can be layered on later once the basic means-end relation is measured reliably across models and prompts.

The current results show that even this coarse taxonomy can expose meaningful differences. Direct answering misses

several blocks. The full wrapper catches blocks but overwarns on controls. Generic conflict check balances the two better in this evaluation. The no-feasible-alternative ablation suggests that the wrapper’s remediation requirement may be the wrong place to add structure.

This makes GICN a useful intermediate benchmark. It is more specific than broad instruction following, less complex than full interactive assistance, and directly tied to a failure mode that users can encounter in ordinary requests. Its role is not to replace existing benchmarks, but to add a missing relation-level check to the evaluation toolbox.